1. AIM

To design a simple Library Management System using HTML, CSS and JavaScript that allows students to search books, manage their borrowing history, check fines and download study materials, while the librarian can manage books, issue and return books, calculate fines and manage study materials.

2. SOURCE CODE

The project uses three files. Keep index.html, style.css and script.js in the same folder.

2.1 HTML — index.html

File: index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Campus Library | Library Management System</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
<header class="topbar">
  <a class="logo" href="#" data-view="home">📚Campus Library</a>
  <button class="menu-button" id="menuButton" aria-label="Open navigation">≡</button>
  <nav id="nav">
    <a href="#" data-view="home">Home</a>
    <a href="#" data-view="books">Books</a>
    <a href="#" data-view="materials">Study Materials</a>
    <a href="#" data-view="contact">Contact</a>
    <button id="loginButton" class="button primary">Login</button>
    <button id="logoutButton" class="button outline hidden">Logout</button>
  </nav>
</header>

<main>
  <section id="homeView" class="view">
    <div class="hero">
      <div class="hero-text">
        <p class="eyebrow">READ · LEARN · GROW</p>
        <h1>Your campus library,<br><span>all in one place.</span></h1>
        <p>Search books, keep track of borrowed items, check fines and download study materials.</p>
        <button class="button primary" data-view="books">Explore books →</button>
      </div>
    </div>
    <section class="search-box container">
      <label for="homeSearch">Find a book</label>
      <div class="search-row"><input id="homeSearch" placeholder="Search by title, author or subject"><button class="button
primary" id="homeSearchButton">Search</button></div>
    </section>
    <section class="container section">
      <div class="section-head"><div><p class="eyebrow">LIBRARY COLLECTION</p><h2>Featured books</h2></div><a href="#" data-
view="books">View all books →</a></div>
      <div id="featuredBooks" class="book-grid"></div>
    </section>
    <section class="info-strip"><div class="container info-grid">
      <article><span>01</span><h3>Search easily</h3><p>Find books by title, author or subject.</p></article>
      <article><span>02</span><h3>Track borrowing</h3><p>Check issue dates, due dates and fines.</p></article>
      <article><span>03</span><h3>Learn anywhere</h3><p>Browse study material by year and semester.</p></article>
    </div></section>
  </section>

  <section id="booksView" class="view hidden container section">
    <p class="eyebrow">LIBRARY CATALOG</p><h1>Search books</h1>
    <p class="muted">Search by book title, author or subject. Select a book to see details.</p>
    <div class="search-row catalog-search"><input id="bookSearch" placeholder="e.g. Data Structures, C Programming"><button
id="bookSearchButton" class="button primary">Search</button></div>
    <div id="bookList" class="book-grid"></div>
    <div id="bookDetail" class="panel hidden"></div>
  </section>

  <section id="dashboardView" class="view hidden container section">
    <p class="eyebrow">STUDENT AREA</p><h1>My dashboard</h1>
    <div id="profileBox" class="panel"></div>
    <div class="section-head"><h2>My borrowing history</h2><button class="button outline" id="refreshHistory">Refresh</button></
div>
    <div id="myLoans"></div>
    <div class="section-head"><h2>My fines</h2></div><div id="myFines"></div>
  </section>
```

```
  <section id="materialsView" class="view hidden container section">
    <p class="eyebrow">LEARNING RESOURCES</p><h1>Study materials</h1>
    <p class="muted">Filter resources by year and semester. Sample files can be downloaded.</p>
    <div class="filter-row">
      <label>Year<select id="materialYear"><optio
n value="all">All years</option><option>1</option><option>2</option><option>3</option><option>4</option></select></label>
      <label>Semester<select id="materialSemester"><option value="all">All semesters</option><option>1</option><option>2</
option><option>3</option><option>4</option><option>5</option><option>6</option><option>7</option><option>8</option></select></
label>
    </div>
    <div id="materialList" class="material-list"></div>
  </section>

  <section id="adminView" class="view hidden container section">
    <p class="eyebrow">LIBRARIAN WORKSPACE</p><h1>Admin dashboard</h1>
    <div class="tabs">
      <button class="tab active" data-admin-tab="books">Manage books</button>
      <button class="tab" data-admin-tab="loans">Issue / Return</button>
      <button class="tab" data-admin-tab="materials">Study materials</button>
      <button class="tab" data-admin-tab="users">Users</button>
    </div>
    <div id="adminContent"></div>
  </section>

  <section id="contactView" class="view hidden container section">
    <p class="eyebrow">GET IN TOUCH</p><h1>Contact the library</h1>
    <div class="two-col">
      <div class="panel"><h2>Library help desk</h2><p>Email: library@campus.example</p><p>Phone: +91 44 5555 0101</p><p>Opening
hours: Monday–Saturday, 9:00 AM–5:00 PM</p></div>
      <form id="contactForm" class="panel form-stack"><h2>Send a message</h2><label>Your name<input id="contactName" required></
label><label>Email<input id="contactEmail" type="email" required></label><label>Message<textarea id="contactMessage" rows="4"
required></textarea></label><button class="button primary">Submit message</button><p class="muted small">Demo form: message is
confirmed in the browser and is not emailed.</p></form>
    </div>
  </section>
</main>

<footer><div class="container footer-row"><div><b>📚 Campus Library</b><p>Simple library management for students and librarians.</
p></div><span>Internet Programming Lab Experiment</span></div></footer>

<dialog id="authDialog" class="dialog">
  <button class="close" data-close="authDialog">×</button>
  <div class="tabs"><button class="tab active" data-mode="login">Login</button><button class="tab" data-mode="register">Register</
button></div>
  <form id="authForm" class="form-stack">
    <h2 id="authTitle">Welcome back</h2>
    <label id="authNameLabel" class="hidden">Full name<input id="authName"></label>
    <label>Email<input id="authEmail" type="email" required></label>
    <label>Password<input id="authPassword" type="password" minlength="4" required></label>
    <p id="authError" class="error"></p>
    <button class="button primary" id="authSubmit">Login</button>
    <p class="small muted">Demo admin: librarian@campus.demo / Library123!</p>
  </form>
</dialog>
<div id="toast" role="status" class="toast"></div>
<script src="script.js"></script>
</body>
</html>
```

2.2 CSS — style.css

File: style.css

```
:root{--ink:#1d302d;--green:#176b57;--green-dark:#104b3f;--pale:#e8f2eb;--bg:#f7f9f6;--muted:#687873;--line:#dce6df;--white:#fff;--
gold:#e8bd67;font-family:Arial,Helvetica,sans-serif;color:var(--ink);background:var(--bg)}*{box-sizing:border-box}body{margin:0}
a{color:var(--green);text-decoration:none}button,input,select,textarea{font:inherit}button{cursor:pointer}.hidden{display:none!
important}.container{width:min(1120px,calc(100% - 36px));margin:auto}.topbar{height:70px;background:#fff;border-bottom:1px solid
var(--line);display:flex;align-items:center;justify-content:space-between;padding:0 max(18px,calc((100vw - 1120px)/
2));position:sticky;top:0;z-index:10}.logo{font-size:1.15rem;font-weight:800;color:var(--ink)}nav{display:flex;align-
items:center;gap:20px;font-size:.9rem}nav a{color:#52635d}nav a:hover{color:var(--green)}.menu-
button{display:none;border:0;background:none;font-size:1.5rem}.button{padding:11px 16px;border:0;border-radius:8px;font-
weight:700}.primary{background:var(--green);color:white}.primary:hover{background:var(--green-
dark)}.outline{background:#fff;border:1px solid #c8d7ce;color:var(--ink)}.hero{min-height:390px;background:linear-
gradient(90deg,#103e33e8,#103e3355),url('https://images.unsplash.com/photo-1507842217343-583bb7270b66?
auto=format&fit=crop&w=1700&q=80') center/cover;display:flex;align-items:center;color:white}.hero-text{width:min(1120px,calc(100% -
36px));margin:auto;padding:50px 0}.eyebrow{font-size:.72rem;letter-spacing:.16em;font-weight:800;color:#77b89d;margin:0 0
12px}.hero h1{font-size:clamp(2.5rem,6vw,4.7rem);line-height:1.02;letter-spacing:-.055em;margin:0}.hero h1
span{color:#d6edb5}.hero-text>p:not(.eyebrow){max-width:500px;line-height:1.7;color:#e2ece6;margin:22px 0}.search-
box{background:white;padding:20px 24px;border-radius:13px;box-shadow:0 12px 35px #102d2217;margin-
top:-25px;position:relative}.search-box label{font-weight:700;display:block;margin-bottom:10px}.search-
row{display:flex;gap:10px}.search-row input{flex:1;min-width:0}input,select,textarea{border:1px solid #cedbd2;border-
radius:8px;padding:12px;background:white;color:var(--ink);width:100%;outline:none}input:focus,select:focus,textarea:focus{border-
color:var(--green);box-shadow:0 0 0 3px #176b5717}.section{padding-top:48px;padding-bottom:60px}.section h1{font-
size:clamp(2rem,4vw,3rem);letter-spacing:-.045em;margin:0 0 12px}.section h2{font-size:1.35rem;margin:0 0 14px}.section-
head{display:flex;justify-content:space-between;align-items:center;gap:12px;margin:10px 0 20px}.section-head
h2{margin:0}.muted{color:var(--muted);line-height:1.6}.small{font-size:.8rem}.book-grid{display:grid;grid-template-
```

```
columns:repeat(4,minmax(0,1fr));gap:16px}.book-card{background:white;border:1px solid var(--line);border-
radius:12px;overflow:hidden;display:flex;flex-direction:column}.book-cover{height:145px;display:grid;place-
items:center;background:linear-gradient(135deg,#d9e8dd,#f1e8d6);font-size:3rem}.book-info{padding:15px;display:flex;flex-
direction:column;gap:7px;flex:1}.book-info h3{font-size:1rem;margin:0}.book-info p{font-size:.82rem;color:var(--
muted);margin:0;line-height:1.5}.book-info .button{margin-top:auto;align-self:start}.info-strip{background:var(--pale);padding:32px
0}.info-grid{display:grid;grid-template-columns:repeat(3,1fr);gap:25px}.info-grid article span{color:var(--green);font-
weight:800}.info-grid h3{margin:8px 0}.info-grid p{color:var(--muted);margin:0;font-size:.9rem;line-height:1.5}.catalog-search{max-
width:650px;margin:20p
x 0 25px}.panel{background:white;border:1px solid var(--line);border-radius:12px;padding:22px;margin:18px 0}.panel h2{margin-
top:0}.book-detail-grid{display:grid;grid-template-columns:140px 1fr;gap:20px}.book-detail-cover{background:var(--pale);border-
radius:10px;display:grid;place-items:center;font-size:3.5rem;min-height:170px}.detail-actions{display:flex;gap:8px;flex-
wrap:wrap;margin-top:15px}.tabs{display:flex;gap:5px;overflow:auto;border-bottom:1px solid var(--line);margin:22px
0}.tab{border:0;background:none;padding:12px 14px;color:var(--muted);font-weight:700;border-bottom:2px solid transparent;white-
space:nowrap}.tab.active{color:var(--green);border-color:var(--green)}.filter-row{display:flex;gap:14px;max-width:500px;margin:20px
0}.filter-row label{flex:1;display:grid;gap:7px;font-size:.85rem;font-weight:700}.material-list{display:grid;gap:12px}.material-
card{background:white;border:1px solid var(--line);border-radius:10px;padding:16px;display:flex;align-items:center;justify-
content:space-between;gap:18px}.material-card h3{margin:0 0 5px;font-size:1rem}.material-card p{margin:0;color:var(--muted);font-
size:.85rem}.form-stack{display:grid;gap:14px}.form-stack label{display:grid;gap:7px;font-size:.85rem;font-weight:700}.form-stack
h2{margin-bottom:2px}.two-col{display:grid;grid-template-columns:1fr 1fr;gap:18px}.two-col .panel{margin:10px 0}.table-
wrap{overflow:auto;border:1px solid var(--line);border-radius:10px;background:#fff;margin:15px 0}table{width:100%;border-
collapse:collapse;min-width:650px}th,td{text-align:left;padding:12px;border-bottom:1px solid var(--line);font-size:.84rem}
th{background:#edf4ef;color:#52635d;font-size:.75rem;text-transform:uppercase}td input,td select{padding:8px;min-width:75px}.admin-
form{display:grid;grid-template-columns:repeat(3,minmax(0,1fr));gap:12px;background:#fff;border:1px solid var(--line);border-
radius:10px;padding:18px;margin:15px 0}.admin-form label{display:grid;gap:6px;font-size:.82rem;font-weight:700}.admin-
form .wide{grid-column:1/-1}.loan-card{background:white;border:1px solid var(--line);border-radius:10px;padding:16px;margin:10px
0;display:grid;grid-template-columns:1.4fr 1fr 1fr auto;gap:12px;align-items:center}.loan-card h3{font-size:.98rem;margin:0 0
5px}.loan-card p{font-size:.82rem;color:var(--muted);margin:3px 0}.status{display:inline-block;border-radius:20px;padding:5px
9px;font-size:.73rem;font-
weight:700;background:#e2f3e7;color:#276c3d}.status.returned{background:#e9eef4;color:#455d74}.status.overdue{background:#fce8e3;co
lor:#a33c2c}.error{color:#b33c30;font-size:.84rem;min-height:1em}.dialog{border:0;border-
radius:14px;padding:28px;width:min(420px,calc(100% - 24px));box-shadow:0 25px 80px
#12352b55}.dialog::backdrop{background:#0d2d2599}.close{position:absolute;right:14px;top:8px;font-
size:1.7rem;border:0;background:none;color:var(--
muted)}.toast{position:fixed;bottom:20px;right:20px;background:#153f34;color:#fff;padding:13px 18px;border-
radius:9px;opacity:0;transform:translateY(10px);transition:.2s;pointer-events:none;z-index:30;max-width:calc(100% -
40px)}.toast.show{opacity:1;transform:translateY(0)}footer{background:#123d32;color:white;padding:28px 0}.footer-
row{display:flex;justify-content:space-between;align-items:center;gap:20px}.footer-row p,.footer-row span{font-
size:.82rem;color:#bad0c5}.empty{background:white;border:1px dashed #c7d7cd;border-radius:10px;padding:25px;text-
align:center;color:var(--muted)}.danger{color:#a33c2c}.small-btn{padding:7px 10px;font-size
:.78rem}
@media(max-width:900px){.book-grid{grid-template-columns:repeat(2,minmax(0,1fr))}nav{gap:12px}.loan-card{grid-template-columns:1fr
1fr}.loan-card>div:last-child{grid-column:1/-1}.admin-form{grid-template-columns:repeat(2,minmax(0,1fr))}}
@media(max-width:640px){.topbar{height:62px}.menu-button{display:block}
nav{display:none;position:absolute;top:61px;left:0;right:0;background:white;padding:18px;box-shadow:0 10px 25px #0001;flex-
direction:column;align-items:stretch}nav.open{display:flex}nav .button{width:100%}.hero{min-height:350px}.hero-text{padding:35px
0}.search-box{padding:17px;margin-top:-15px}.section{padding-top:36px;padding-bottom:40px}.book-grid{grid-template-columns:1fr
1fr;gap:10px}.book-cover{height:110px}.book-info{padding:11px}.book-info h3{font-size:.92rem}.book-info .button{padding:9px
10px;font-size:.8rem}.info-grid,.two-col{grid-template-columns:1fr}.info-grid{gap:20px}.filter-row{flex-direction:column}.material-
card{align-items:flex-start;flex-direction:column}.admin-form{grid-template-columns:1fr}.admin-form .wide{grid-column:auto}.loan-
card{grid-template-columns:1fr}.loan-card>div:last-child{grid-column:auto}.footer-row{align-items:flex-start;flex-
direction:column}.book-detail-grid{grid-template-columns:1fr}.book-detail-cover{min-height:110px}.container{width:calc(100% -
28px)}}
```

2.3 JavaScript — script.js

File: script.js

```
/* Campus Library: simple front-end lab project.
   Demo data is saved in localStorage. Use fictional details only. */
"use strict";
const KEY="campus_library_demo_v1";
const ADMIN_EMAIL="librarian@campus.demo";
const ADMIN_PASSWORD="Library123!";
const isoToday=()=>{const d=new Date();d.setMinutes(d.getMinutes()-d.getTimezoneOffset());return d.toISOString().slice(0,10)};
const addDays=(date,n)=>{const d=new Date(date+"T12:00:00");d.setDate(d.getDate()+n);return d.toISOString().slice(0,10)};
const money=n=>"₹"+Math.round(Number(n)||0).toLocaleString("en-IN");
const esc=s=>String(s??"").replace(/[&<>"']/g,c=>({"&":"&amp;","<":"&lt;",">":"&gt;","'":"&quot;","'":"&#39;"}[c]));
const dayDiff=(a,b)=>Math.round((new Date(b+"T12:00:00")-new Date(a+"T12:00:00"))/86400000);
const starterBooks=[
 {id:1,title:"Data Structures",author:"Seymour Lipschutz",subject:"Computer Science",year:2022,copies:5,available:5,description:"A
practical introduction to lists, stacks, queues, trees and graphs."},
 {id:2,title:"Programming in C",author:"E.
Balagurusamy",subject:"Programming",year:2021,copies:4,available:4,description:"Fundamentals of C programming, control flow,
functions and arrays."},
 {id:3,title:"Database System Concepts",author:"Abraham
Silberschatz",subject:"Database",year:2020,copies:3,available:3,description:"Relational databases, SQL, transactions and database
design."},
 {id:4,title:"Computer Networks",author:"Andrew S.
Tanenbaum",subject:"Networking",year:2023,copies:4,available:4,description:"Network architectures, protocols, routing and data
communication."},
 {id:5,title:"Operating System Concepts",author:"Abraham Silberschatz",subject:"Operating
Systems",year:2022,copies:3,available:3,description:"Processes, memory management, file systems and scheduling."},
 {id:6,title:"Web Technology",author:"Uttam K. Roy",subject:"Web Development",year:2021,copies:2,available:2,description:"Core
concepts used to build websites and web applications."}
];
const starterMaterials=[
```

```
 {id:1,title:"C Programming Notes",subject:"Programming",year:"1",semester:"1",fileName:"c-programming-notes.txt",content:"C
PROGRAMMING NOTES\n\n1. A program is a set of instructions.\n2. Common data types include int, char, float and double.\n3. Control
statements include if, switch, for, while and do-while.\n4. Functions help divide a program into reusable units.\n"},
 {id:2,title:"Data Structures Revision",subject:"Computer Science",year:"2",semester:"3",fileName:"data-structures-
notes.txt",content:"DATA STRUCTURES REVISION\n\nStack: LIFO structure.\nQueue: FIFO structure.\nLinked list: nodes connected by
links.\nTree: hierarchical data structure.\nGraph: vertices connected by edges.\n"},
 {id:3,title:"Database SQL Basics",subject:"Database",year:"2",semester:"4",fileName:"database-sql-basics.txt",content:"DATABASE
SQL BASICS\n\nSELECT * FROM Students;\nINSERT INTO Students(id, name) VALUES (1, 'Example');\nUPDATE Students SET name='New Name'
WHERE id=1;\nDELETE FROM Students WHERE id=1;\n"},
 {id:4,title:"Computer Networks Summary",subject:"Networking",year:"3",semester:"5",fileName:"network-
summary.txt",content:"COMPUTER NETWORKS SUMMARY\n\nOSI layers: Physical, Data Link, Network, Transport, Session, Presentation,
Application.\nTCP provides reliable, connection-oriented transport.\nIP provides logical addressing and packet routing.\n"}
];
function fresh(){return {books:structuredClone(starterBooks),users:[],loans:
[],materials:structuredClone(starterMaterials),messages:
[],nextBook:7,nextLoan:1001,nextMaterial:5,currentUserId:null};}
function load(){try{const x=JSON.parse(localStorage.getItem(KEY));if(x)return {...fresh(),...x};}catch{}return fresh();}
let db=load(),authMode="login",adminTab="books",selectedBook=null;
function save(){localStorage.setItem(KEY,JSON.stringify(db))}
function user(){return db.users.find(u=>u.id===db.currentUserId)||null}
function admin(){return user()?.role==="admin"}
function toast(msg){const
t=document.getElementById("toast");t.textContent=msg;t.classList.add("show");clearTimeout(toast.timer);toast.timer=setTimeout(()=>t
.classList.remove("show"),2800)}
function view(name)
{document.querySelectorAll(".view").forEach(v=>v.classList.add("hidden"));document.getElementById(name+"View").classList.remove("hi
dden");document.getElementById("nav").classList.remove("open");if(name==="books")renderBooks();if(name==="materials")renderMaterial
s();if(name==="dashboard")renderDashboard();if(name==="admin")renderAdmin();window.scrollTo({top:0,behavior:"smooth"})}
function syncNav(){const u=user();document.getElementById("loginButton").classList.toggle("hidden",!!
u);document.getElementById("logoutButton").classList.toggle("hidden",!u);document.querySelector('[data-
view="materials"]').classList.remove("hidden")}
function showAuth(mode="login")
{authMode=mode;syncAuthMode();document.getElementById("authError").textContent="";document.getElementById("authDialog").showModal()
}
function syncAuthMode(){document.getElementById("authTitle").textContent=authMode==="login"?"Welcome back":"Create student
account";document.getElementById("authNameLabel").classList.toggle("hidden",authMode!
=="register");document.getElementById("authName").required=authMode==="register";document.getElementById("authSubmit").textContent=
authMode==="login"?"Login":"Register";document.querySelectorAll("[data-
mode]").forEach(b=>b.classList.toggle("active",b.dataset.mode===authMode))}
function authSubmit(e){e.preventDefault();const
email=document.getElementById("authEmail").value.trim().toLowerCase(),password=document.getElementById("authPassword").value,err=do
cument.getElementById("authError");
 if(email===ADMIN_EMAIL&&password===ADMIN_PASSWORD){let a=db.users.find(u=>u.email===ADMIN_EMAIL);if(!a)
{a={id:"admin",name:"Librarian Admin",email,password,role:"admin",phone:""};db.users.push(a)}
db.currentUserId=a.id;save();document.getElementById("authDialog").close();syncNav();view("admin");toast("Admin login
successful.");return}
 if(authMode==="register"){const name=document.getElementById("authName").value.trim();if(name.length<2){err.textContent="Enter
your name.";return}if(db.users.some(u=>u.email===email)){err.textContent="This email is already registered.";return}const
u={id:"u"+Date.now(),name,email,password,role:"student",phone:""};db.users.push(u);db.currentUserId=u.id;save();document.getElement
ById("authDialog").close();syncNav();view("dashboard");toast("Registration successful.");}
 else{const u=db.users.find(u=>u.email===email&&u.password===password&&u.role!=="admin");if(!u){err.textContent="Invalid email or
password.";return}
db.currentUserId=u.id;save();document.getElementById("authDialog").close();syncNav();view("dashboard");toast("Login successful.")}}
function logout(){db.currentUserId=null;save();syncNav();view("home");toast("Logged out.")}
function bookCard(b){return `<article class="book-card"><div class="book-cover">📘</div><div class="book-info"><h3>${esc(b.title)}
</h3><p>By ${esc(b.author)}</p><p>${esc(b.subje
ct)} · ${b.year}</p><p><b>${b.available}</b> of ${b.copies} available</p><button class="button outline small-btn" data-book-
detail="${b.id}">View details</button></div></article>`}
function renderHomeBooks(){document.getElementById("featuredBooks").innerHTML=db.books.slice(0,4).map(bookCard).join("")}
function renderBooks(){const q=(document.getElementById("bookSearch")?.value||"").trim().toLowerCase();const
found=db.books.filter(b=>[b.title,b.author,b.subject].some(v=>v.toLowerCase().includes(q)));document.getElementById("bookList").inn
erHTML=found.length?found.map(bookCard).join(""):'<div class="empty">No books match your search.</
div>';document.getElementById("bookDetail").classList.add("hidden")}
function detailBook(id){const b=db.books.find(b=>b.id===Number(id));if(!b)return;selectedBook=b.id;const
el=document.getElementById("bookDetail");el.classList.remove("hidden");el.innerHTML=`<div class="book-detail-grid"><div
class="book-detail-cover">📚</div><div><p class="eyebrow">BOOK DETAILS</p><h2>${esc(b.title)}</h2><p><b>Author:</b> $
{esc(b.author)}</p><p><b>Subject:</b> ${esc(b.subject)}</p><p><b>Publication year:</b> ${b.year}</p><p><b>Availability:</b> $
{b.available} / ${b.copies}</p><p>${esc(b.description)}</p><div class="detail-actions">${user()&&!admin()?`<button class="button
primary" data-request-issue="${b.id}" ${b.available<1?"disabled":""}>Request issue</button>`:`<button class="button primary" data-
require-login>Login to borrow</button>`}<button class="button outline" data-close-detail>Close details</button></div></div></
div>`;el.scrollIntoView({behavior:"smooth",block:"nearest"})}
function requestIssue(id){const u=user();if(!u||admin()){showAuth("login");return}const b=db.books.find(b=>b.id===Number(id));if(!
b||b.available<1){toast("No copies are currently available.");return}
if(db.loans.some(l=>l.bookId===b.id&&l.userId===u.id&&l.status==="Issued")){toast("You already have this book issued.");return}
const loan={id:String(db.nextLoan+
+),bookId:b.id,userId:u.id,issueDate:isoToday(),dueDate:addDays(isoToday(),14),returnDate:"",status:"Issued",fine:0};db.loans.push(
loan);b.available--;save();renderBooks();renderDashboard();toast("Book issued in demo. Due date: "+loan.dueDate)}
function fineFor(loan){if(loan.status==="Returned")return Number(loan.fine)||0;return
Math.max(0,dayDiff(loan.dueDate,isoToday()))*5}
function renderDashboard(){const u=user();if(!u){view("home");showAuth();return}
document.getElementById("profileBox").innerHTML=`<h2>My profile</h2><form id="profileForm" class="admin-form"><label>Name<input
id="profileName" value="${esc(u.name)}" required></label><label>Email<input value="${esc(u.email)}" disabled></
label><label>Phone<input id="profilePhone" value="${esc(u.phone||"")}" placeholder="Phone number"></label><button class="button
primary">Update profile</button></form>`;

document.getElementById("profileForm").addEventListener("submit",e=>{e.preventDefault();u.name=document.getElementById("profileName
").value.trim();u.phone=document.getElementById("profilePhone").value.trim();save();toast("Profile updated.")});
 const
```

```
loans=db.loans.filter(l=>l.userId===u.id).sort((a,b)=>b.issueDate.localeCompare(a.issueDate));document.getElementById("myLoans").innerHTML=loans.length?loans.map(l=>{const b=db.books.find(b=>b.id===l.bookId)||{title:"Deleted book"};const f=fineFor(l);return `<article class="loan-card"><div><h3>${esc(b.title)}</h3><p>Loan ID: ${l.id}</p><p>Issue date: ${l.issueDate}</p><p>Expected return: ${l.due
Date}</p>${l.returnDate?`<p>Returned: ${l.returnDate}</p>`:""}</div><div><span class="status ${l.status==="Returned"?"returned":f?"overdue":""}">${l.status}</span></div><div><b>Fine: ${money(f)}</b><p>${f?"₹5 per overdue day":"No fine currently"}</p></div><div>${l.status==="Issued"?"<span class='small muted'>Return via librarian</span>":""}</div></article>`}).join(""):'<div class="empty">You have no borrowing history yet. Search books to get started.</div>';
 const total=loans.reduce((s,l)=>s+fineFor(l),0);document.getElementById("myFines").innerHTML=`<div class="panel"><h2>Total fine: ${money(total)}</h2><p class="muted small">Demo rule: ₹5 per day after the expected return date. Actual payment is not processed by this website.</p></div>`}
function renderMaterials(){const y=document.getElementById("materialYear").value,s=document.getElementById("materialSemester").value;const list=db.materials.filter(m=>(y==="all"||m.year===y)&&(s==="all"||m.semester===s));document.getElementById("materialList").innerHTML=list.length?list.map(m=>`<article class="material-card"><div><h3>${esc(m.title)}</h3><p>${esc(m.subject)} · Year ${m.year} · Semester ${m.semester}</p></div><button class="button primary" data-download-material="${m.id}">Download</button></article>`).join(""):'<div class="empty">No materials for this year and semester.</div>'}
function downloadMaterial(id){const m=db.materials.find(m=>m.id===Number(id));if(!m)return;const blob=new Blob([m.content||`${m.title}\n${m.subject}\nYear ${m.year}, Semester ${m.semester}`],{type:"text/plain"});const url=URL.createObjectURL(blob);const a=document.createElement("a");a.href=url;a.download=m.fileName||"study-material.txt";a.click();URL.revokeObjectURL(url);toast("Study material download started.")}
function renderAdmin(){if(!admin()){view("home");showAuth();return}document.querySelectorAll("[data-admin-tab]").forEach(b=>b.classList.toggle("active",b.dataset.adminTab===adminTab));const el=document.getElementById("adminContent");
 if(adminTab==="books"){el.innerHTML=`<h2>Book inventory</h2><form id="addBookForm" class="admin-form"><label>Book title<input id="newTitle" required></label><label>Author<input id="newAuthor" required></label><label>Subject<input id="newSubject" required></label><label>Publication year<input id="newYear" type="number" min="1900" max="2100" value="2024" required></label><label>Total copies<input id="newCopies" type="number" min="1" value="1" required></label><label class="wide">Description<input id="newDescription" required></label><button class="button primary">Add book</button></form><div class="table-wrap"><table><thead><tr><th>Book</th><th>Author / Subject</th><th>Year</th><th>Total</th><th>Available</th><th>Actions</th></tr></thead><tbody>${db.books.map(b=>`<tr><td>${esc(b.title)}</td><td>${esc(b.author)}<br>${esc(b.subject)}</td><td>${b.year}</td><td><input type="number" min="${b.copies-b.available}" value="${b.copies}" id="copies-${b.id}"></td><td>${b.available}</td><td><button class="button outline small-btn" data-update-book="${b.id}">Update</button> <button class="button outline small-btn danger" data-delete-book="${b.id}">Delete</button></td></tr>`).join("")}</tbody></table></div>`;
 document.getElementById("addBookForm").addEventListener("submit",e=>{e.preventDefault();const copies=Number(document.getElementById("newCopies").value);db.books.push({id:db.nextBook++,title:document.getElementById("newTitle").value.trim(),author:document.getElementById("newAuthor").value.trim(),subject:document.getElementById("newSubject").value.trim(),year:Number(document.getElementById("newYear").value),copies,available:copies,description:document.getElementById("newDescription").value.trim()});save();renderAdmin();renderHomeBooks();toast("Book added.")});
 }else if(adminTab==="loans"){el.innerHTML=`<h2>Issue / return books</h2><p class="muted small">Student must be registered. Issue a book by choosing a student and an available title.</p><form id="issueForm" class="admin-form"><label>Student<select id="issueUser">${db.users.filter(u=>u.role==="student").map(u=>`<option value="${u.id}">${esc(u.name)} (${esc(u.email)})</option>`).join("")}</select></label><label>Book<select id="issueBook">${db.books.filter(b=>b.available>0).map(b=>`<option value="${b.id}">${esc(b.title)} — ${b.available} available</option>`).join("")}</select></label><label>Loan period<select id="loanPeriod"><option value="7">7 days</option><option value="14" selected>14 days</option><option value="21">21 days</option></select></label><button class="button primary">Issue book</button></form><div class="table-wrap"><table><thead><tr><th>Student</th><th>Book</th><th>Issue date</th><th>Due date</th><th>Status</th><th>Fine</th><th>Action</th></tr></thead><tbody>${db.loans.map(l=>{const u=db.users.find(u=>u.id===l.userId)||{name:"Unknown"};const b=db.books.find(b=>b.id===l.bookId)||{title:"Deleted book"};return `<tr><td>${esc(u.name)}</td><td>${esc(b.title)}</td><td>${l.issueDate}</td><td>${l.dueDate}</td><td>${l.status}</td><td>${money(fineFor(l))}</td><td>${l.status==="Issued"?`<button class="button primary small-btn" data-return-loan="${l.id}">Mark returned</button>`:"—"}</td></tr>`}).join("")}</tbody></table></div>`;
 document.getElementById("issueForm").addEventListener("submit",e=>{e.preventDefault();const u=db.users.find(u=>u.id===document.getElementById("issueUser").value),b=db.books.find(b=>b.id===Number(document.getElementById("issueBook").value));if(!u||!b||b.available<1){toast("Choose a student and an available book.");return}
if(db.loans.some(l=>l.userId===u.id&&l.bookId===b.id&&l.status==="Issued")){toast("This student already has this book issued.");return}db.loans.push({id:String(db.nextLoan++),bookId:b.id,userId:u.id,issueDate:isoToday(),dueDate:addDays(isoToday(),Number(document.getElementById("loanPeriod").value)),returnDate:"",status:"Issued",fine:0});b.available--;save();renderAdmin();renderHomeBooks();toast("Book issued successfully.")});
 }else if(adminTab==="materials"){el.innerHTML=`<h2>Manage study materials</h2><form id="materialForm" class="admin-form"><label>Material title<input id="matTitle" required></label><label>Subject<input id="matSubject" required></label><label>Year<select id="matYear"><option>1</option><option>2</option><option>3</option><option>4</option></select></label><label>Semester<select id="matSemesterAdmin">${[1,2,3,4,5,6,7,8].map(n=>`<option>${n}</option>`).join("")}</select></label><label class="wide">Text content<input id="matContent" required placeholder="Paste short notes or a resource description"></label><button class="button primary">Add material</button></form><div class="table-wrap"><table><thead><tr><th>Material</th><th>Subject</th><th>Year</th><th>Semester</th><th>Action</th></tr></thead><tbody>${db.materials.map(m=>`<tr><td>${esc(m.title)}</td><td>${esc(m.subject)}</td><td>${m.year}</td><td>${m.semester}</td><td><button class="button outline small-btn danger" data-delete-material="${m.id}">Delete</button></td></tr>`).join("")}</tbody></table></div>`;
 document.getElementById("materialForm").addEventListener("submit",e=>{e.preventDefault();const title=document.getElementById("matTitle").value.trim();db.materials.push({id:db.nextMaterial++,title,subject:document.getElementById("matSubject").value.trim(),year:document.getElementById("matYear").value,semester:document.getElementById("matSemesterAdmin").value,fileName:title.toLowerCase().replace(/[^a-z0-9]+/g,"-")+".txt",content:document.getElementById("matContent").value});save();renderAdmin();toast("Study material added.")});
 }else{el.innerHTML=`<h2>Registered users</h2><div class="table-wrap"><table><thead><tr><th>Name</th><th>Email</th><th>Phone</th><th>Role</th><th>Active loans</th></tr></thead><tbody>${db.users.map(u=>`<tr><td>${esc(u.name)}</td><td>${esc(u.email)}</td><td>${esc(u.phone||"—")}</td><td>${u.role}</td><td>${db.loans.filter(l=>l.userId===u.id&&l.status==="Issued").length}</td></tr>`).join("")}</tbody></table></div>`}
}
function returnLoan(id){const l=db.loans.find(l=>l.id===String(id));if(!l||l.status!=="Issued")return;const b=db.books.find(b=>b.id===l.bookId);l.status="Returned";l.returnDate=isoToday();l.fine=fineFor({...l,status:"Issued"});if(b)b.available=Math.min(b.copies,b.available+1);save();renderAdmin();renderHomeBooks();toast(`Book returned. Fine: ${money(l.fine)}`)}
function updateBook(id){const b=db.books.find(b=>b.id===Number(id));if(!b)return;const copies=Number(document.getElementById("copies-"+id).value);if(!Number.isInteger(copies)||copies<b.copies-b.available){toast("Total copies cannot be less than copies already issued.");return}const issued=b.copies-b.available;b.copies=copies;b.available=copies-issued;save();renderAdmin();renderHomeBooks();toast("Book updated.")}
function deleteBook(id){const b=db.books.find(b=>b.id===Number(id));if(!b)return;if(db.loans.some(l=>l.bookId===b.id&&l.status==="Issued")){toast("Return all issued copies before deleting this
```

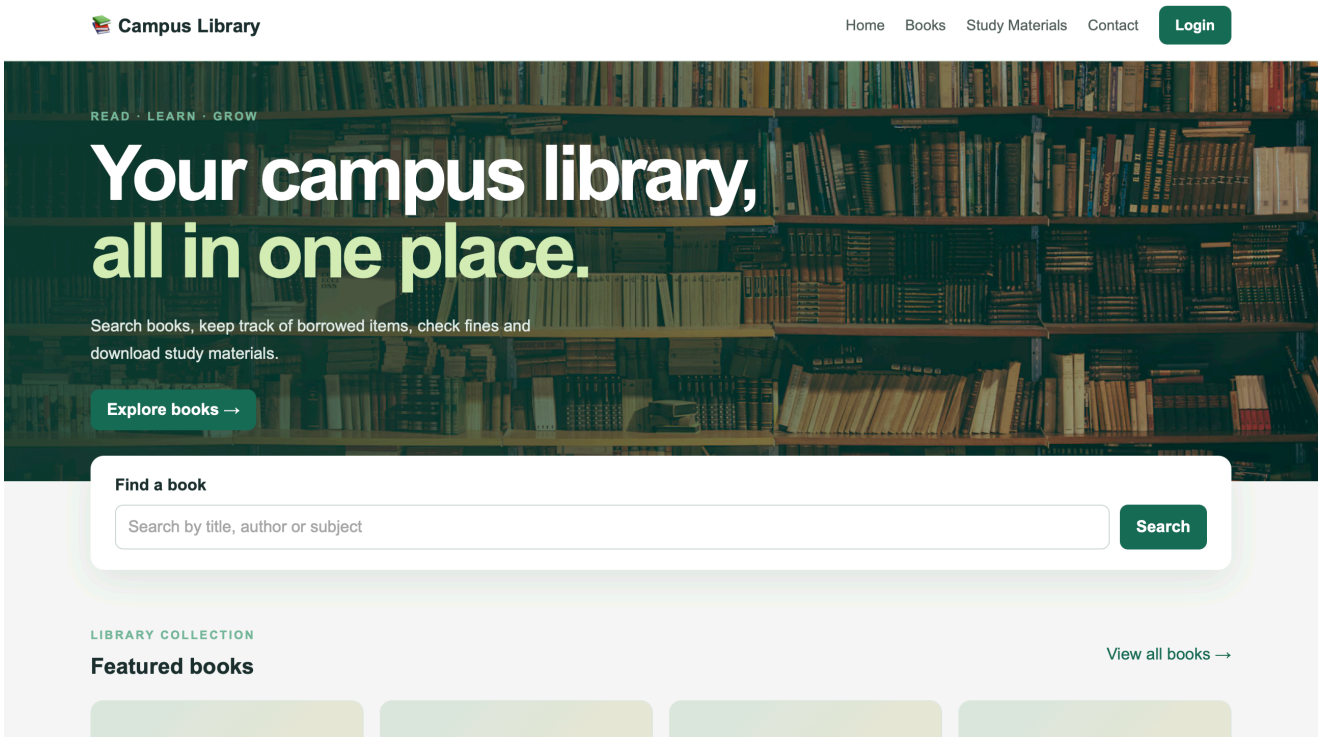
```
book.");return}if(!confirm(`Delete "${b.title}"?`))return;db.books=db.books.filter(x=>x.id!
==b.id);save();renderAdmin();renderHomeBooks();toast("Book deleted.")}
function globalClick(e){const v=e.target.closest("[data-view]");if(v){e.preventDefault();view(v.dataset.view);return}const
d=e.target.closest("[data-book-detail]");if(d){view("books");detailBook(d.dataset.bookDetail);return}const
req=e.target.closest("[data-request-issue]");if(req){requestIssue(req.dataset.requestIssue);return}if(e.target.closest("[data-
require-login]")){showAuth();return}if(e.target.closest("[data-close-detail]"))
{document.getElementById("bookDetail").classList.add("hidden");return}const t=e.target.closest("[data-admin-tab]");if(t)
{adminTab=t.dataset.adminTab;renderAdmin();return}const up=e.target.closest("[data-update-book]");if(up)
{updateBook(up.dataset.updateBook);return}const del=e.target.closest("[data-delete-book]");if(del)
{deleteBook(del.dataset.deleteBook);return}const ret=e.target.closest("[data-return-loan]");if(ret)
{returnLoan(ret.dataset.returnLoan);return}const dm=e.target.closest("[data-delete-material]");if(dm)
{db.materials=db.materials.filter(m=>m.id!==Number(dm.dataset.deleteMaterial));save();renderAdmin();toast("Material
deleted.");return}const dl=e.target.closest("[data-download-material]");if(dl)
{downloadMaterial(dl.dataset.downloadMaterial);return}const mode=e.target.closest("[data-mode]");if(mode)
{authMode=mode.dataset.mode;syncAuthMode();return}const close=e.target.closest("[data-close]");if(c
lose){document.getElementById(close.dataset.close).close()}}
function init(){renderHomeBooks();renderMaterials();syncNav();document.getElementById("loginButton").addEventListener("click",
()=>showAuth());document.getElementById("logoutButton").addEventListener("click",logout);document.getElementById("authForm").addEve
ntListener("submit",authSubmit);document.getElementById("menuButton").addEventListener("click",
()=>document.getElementById("nav").classList.toggle("open"));document.getElementById("homeSearchButton").addEventListener("click",
()=>{document.getElementById("bookSearch").value=document.getElementById("homeSearch").value;view("books");renderBooks()});document
.getElementById("bookSearchButton").addEventListener("click",renderBooks);document.getElementById("bookSearch").addEventListener("i
nput",renderBooks);document.getElementById("materialYear").addEventListener("change",renderMaterials);document.getElementById("mate
rialSemester").addEventListener("change",renderMaterials);document.getElementById("refreshHistory").addEventListener("click",render
Dashboard);document.getElementById("contactForm").addEventListener("submit",e=>{e.preventDefault();db.messages.push({name:document.
getElementById("contactName").value,email:document.getElementById("contactEmail").value,message:document.getElementById("contactMes
sage").value,date:isoToday()});save();e.target.reset();toast("Message saved in this
demo.")});document.addEventListener("click",globalClick)}
init();
```

3. PROCEDURE

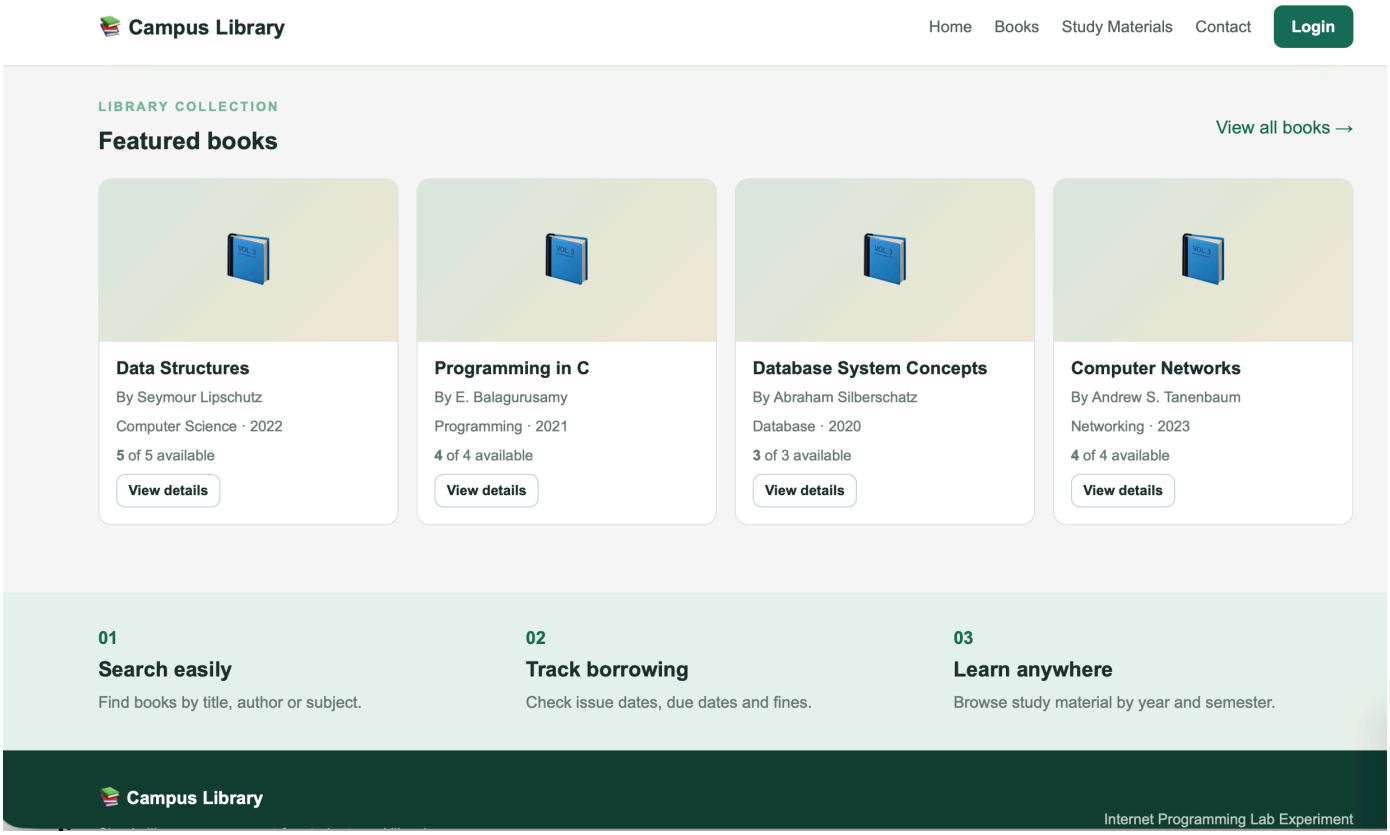
1. Extract Campus_Library_Management_System.zip.
2. Keep index.html, style.css and script.js together in one folder.
3. Open index.html in a browser, or run `python3 -m http.server 8000` from the folder and visit http://localhost:8000.
4. Register a student account, search a book, and test the dashboard and study materials.
5. Log in as librarian to add/update/delete books, issue and return books, calculate fines, and manage study materials.

4. OUTPUT SCREENSHOTS

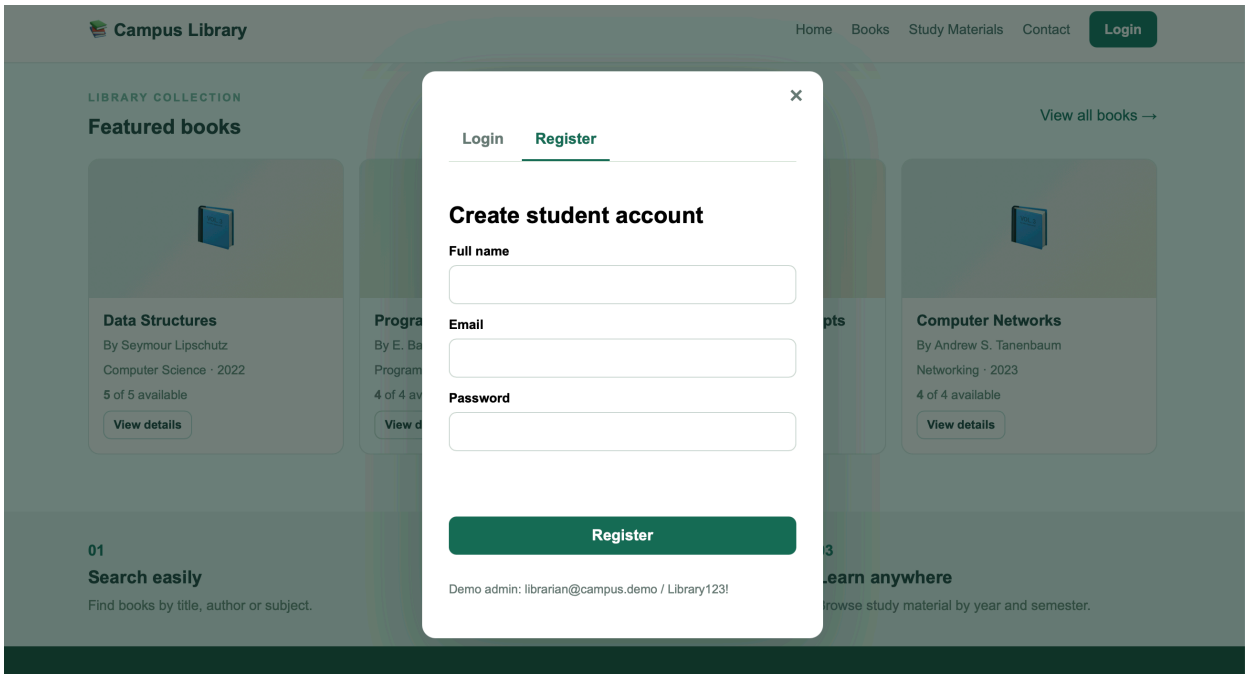
Home page



Book catalog and details



Student registration / login



5. RESULT

The simple Library Management System was created using HTML, CSS and JavaScript. It demonstrates book search, user registration, borrowing history, fine calculation, book inventory management, issue/return processing, study-material downloads and a contact form.

Note

This is a front-end demonstration that stores data in the current browser's localStorage. Login is not production-secure, and the contact form does not send email. Demo librarian login: librarian@campus.demo / Library123!.